

Wednesday Warm Up: Unit 3, Magnetic Forces and Fields

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1 Memory Bank

- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$... The direction of the *cross-product* follows this pattern. Reversing the order of any two vectors introduces a minus sign.
- $\hat{i} \times \hat{i} = 0$, $\hat{j} \times \hat{j} = 0$, $\hat{k} \times \hat{k} = 0$... The *cross-product* is zero when both vectors are parallel.
- $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$... The magnitude of the cross-product of two vectors $\vec{a}$ and $\vec{b}$, given the angle θ between them.
- $\vec{F} = I\vec{L} \times \vec{B}$... The Lorentz force, by a magnetic field $\vec{B}$ on a *current* i of length and direction $\vec{L}$.
- $\vec{F} = q\vec{v} \times \vec{B}$... The Lorentz force, by a magnetic field $\vec{B}$ on a *charge* q with velocity $\vec{v}$.

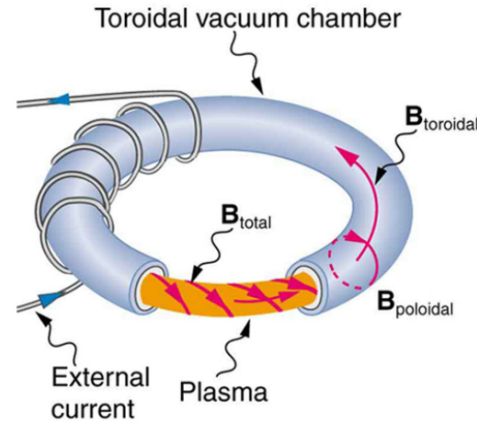


Figure 1: This structure is called a *tokamak*, and it is used to confine plasma in a fusion reactor.

2 The Cross-Product

1. Let $\vec{a} = 2\hat{i} - 1\hat{j}$, and $\vec{b} = 4\hat{i} - 2\hat{j}$. Calculate $\vec{a} \times \vec{b}$. What is the angle between the two vectors?

of an electron? (c) Discuss why the ratio found in (b) should be an integer.

3 Magnetic Force on Currents and Charge

1. A cosmic ray electron moves at $7.50 \times 10^6 \text{ m s}^{-1}$ perpendicular to the magnetic field of the Earth at an altitude where field strength is $1.00 \times 10^{-5} \text{ T}$. What is the radius of the circular path the electron follows?
2. (a) An oxygen-16 ion with a mass of $2.66 \times 10^{-26} \text{ kg}$ travels at $5.00 \times 10^6 \text{ m s}^{-1}$ perpendicular to a 1.20 T magnetic field, which makes it move in a circular arc with a 0.231 m radius. What positive charge is on the ion? (b) What is the ratio of this charge to the charge

3. Consider Fig. 1, in which a *tokamak* is shown. Tokamaks are systems designed to contain a plasma for fusion reactions. (a) Use the right hand rule to show that the solenoidal B-field that we have measured becomes *toroidal*, given the geometry of the external current. (b) The solenoidal and toroidal B-fields originate from Ampère's Law, which states that a current creates a B-field that circles around it. If positive ions in the plasma are circling as shown in Fig. 1, show that they produce a poloidal B-field, as shown. (c) Convince yourself that the sum of the toroidal and poloidal B-fields forms the corkscrew pattern shown in Fig. 1.